Nordic / u-blox / Avnet Seminar

nRF54L hands-on workshop

Nordic Semiconductor
June 2026
Kevin Kotinkar
Regional Sales Manager

Agenda

- **Nordic Product Update**
  - nRF54L series expansion
  - nRF Cloud fixed-rate lifetime FOTA
  - PMIC update

- **Nordic SDK Introduction**
  - Code base and architecture
  - Zephyr project
  - IDE & toolchain overview
  - Learning material

We are Nordic

Simplifying lives through all things connected

Fabless semiconductor company

Founded in Norway, 1983

~1,400 employees

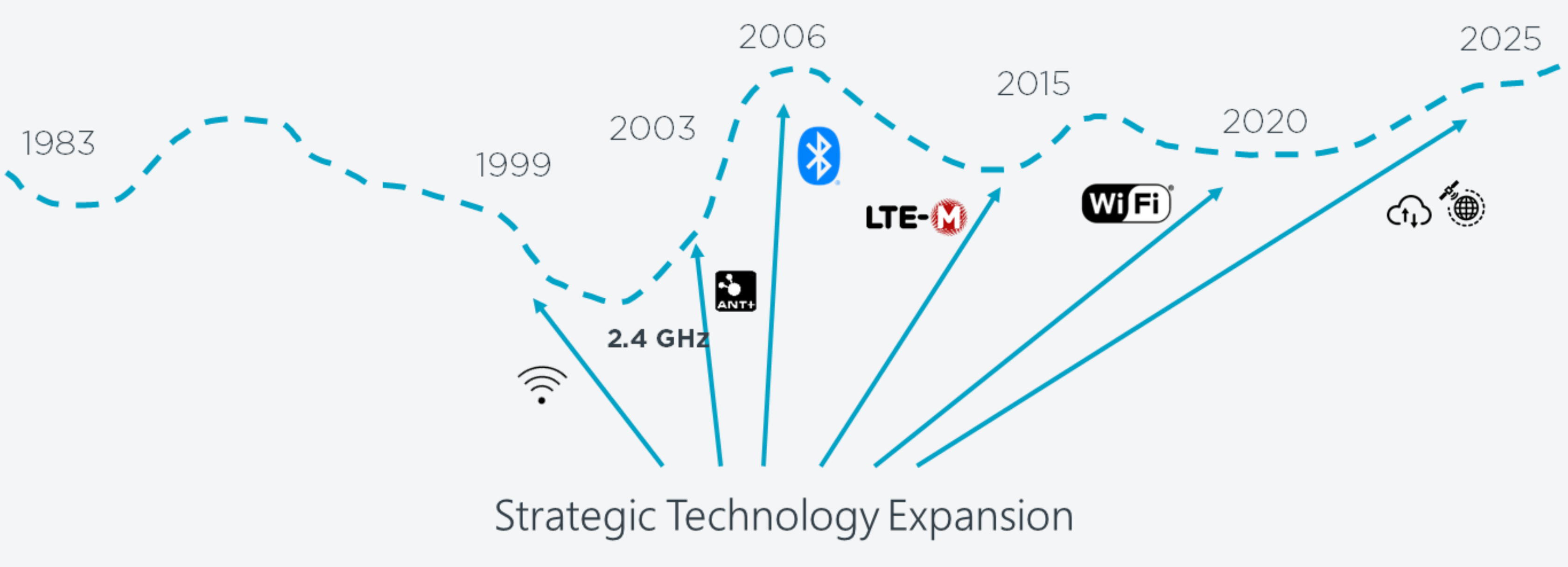
$688 MUSD revenue CY 2025
OSEBX: NOD

ISO 9001 certified

Global presence

Nordic`s technology journey

Wireless connectivity leader



Nordic product overview

Cloud support across all our wireless connectivity solutions

Short-Range	Cellular IoT	Wi-Fi 6 IoT	Power Management
nRF 54 SERIES	nRF 91 SERIES	nRF 70 SERIES	nPM FAMILY
Bluetooth, THREAD, matter, zigbee	LTE-M, NB-IoT, GNSS	WiFi, matter	
nRF CLOUD powered by Memfault	nRF CLOUD powered by Memfault	nRF CLOUD powered by Memfault	

Next-level performance

Successor of the most popular wireless SoCs

nRF52 SERIES

- Shipped in the billions
- Used in ~40% of all Bluetooth LE qualified products from 2017 to 2024
- Valued by engineers for ease of use, performance, and technical support

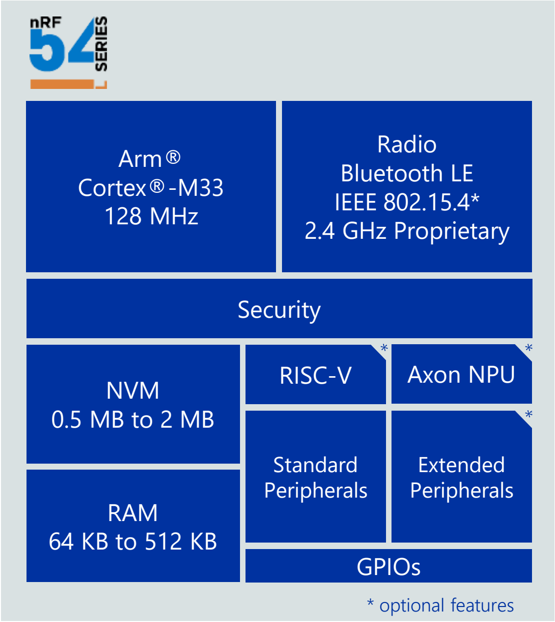
nRF54 L SERIES

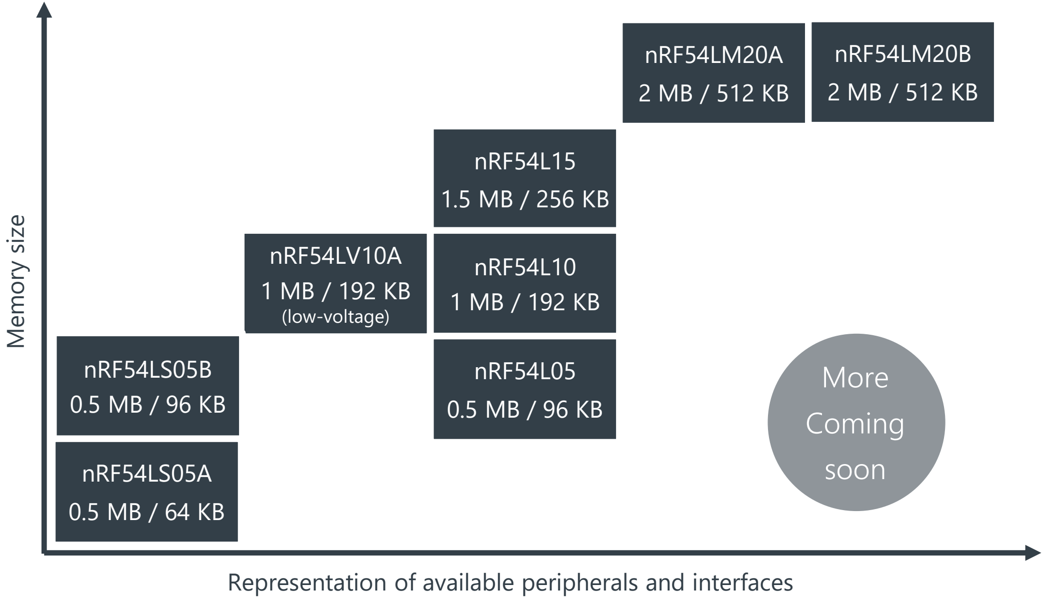
- Bluetooth LE with proven reliability and the latest features
- Multiprotocol with support for Matter and Aliro
- 2x processing power
- 3x processing efficiency
- New 4th gen radio
- 30% to 50% lower power consumption
- Higher memory size options
- New peripherals and interfaces

SoC Series

Overview

Common architecture – scalability and portability





Nordic driving edge AI today

Enhancing intelligent at the edge

Axon	Neuton	EDGE AI LAB
Integrated NPU	Tiny CPU-run models	Development platform

NORDIC SEMICONDUCTOR

EU Cyber Resilience Act (CRA) requirements

Nordics FOTA solution is purpose-built for CRA compliance

nRF CLOUD
powered by Memfault

Vulnerability handling period

The support period for which the manufacturer ensures the effective handling of vulnerabilities should be **no less than five years**, unless the lifetime of the product with digital elements is less than five years...Where the time the product with digital elements is reasonably expected to be in use is longer than five years...manufacturers should accordingly **ensure longer support periods**.

Section 60

Automatic security updates

Ensure that vulnerabilities can be addressed through security updates, including, where applicable, **through automatic security updates** that are installed within an appropriate timeframe.

Annex 1 – Part I, (2)(c)

Secure updates distribution

Provide for mechanisms to securely distribute updates for products with digital elements to ensure that vulnerabilities are fixed or mitigated in a timely manner and, where applicable for security updates, in an automatic manner.

Annex 1 – Part II, (7)

NORDIC SEMICONDUCTOR

Chip-to-cloud solution

A complete FOTA solution right out of the box

Device side done for you

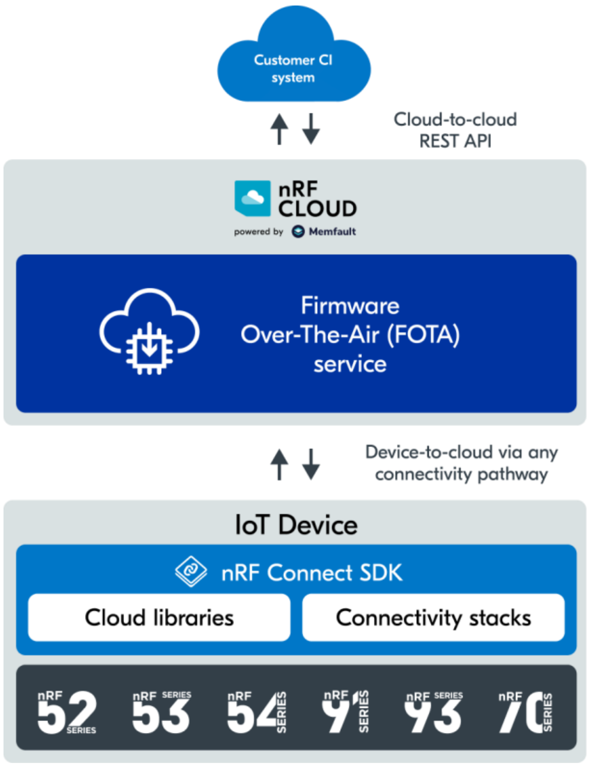
nRF Connect SDK includes everything you need to update devices with no re-engineering required

Integrate cloud in minutes

Connect your device seamlessly to nRF Cloud infrastructure with a few K-configs

Production-ready from day 1

Instant access to a system tried and tested on millions of devices already in the field



FOTA simplified

Controlled, low risk update workflows at scale

Automated rollout management

Highly automated and flexible by allowing you to create custom groups of devices and deploy staged rollouts by group

One-click abort

Should the unforeseen happen we help you mitigate risk by providing “one-click” abort during device update operation

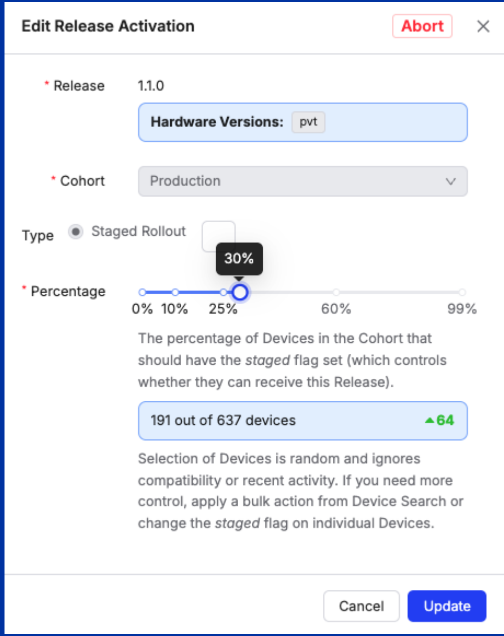
Version control and rollback

Enables version checking and controlled rollback of firmware versions minimizing downtime and maintaining operational continuity

Scale to millions of devices

Our FOTA solution is being successfully used by companies everything from pilot deployments to millions of connected devices

nRF CLOUD
powered by Memfault



NORDIC SEMICONDUCTOR

Fixed-rate lifetime FOTA

Save when you commit

Low cost	Pay one low fixed cost upfront for a lifetime of updates for your devices
Lifetime support	From prototypes to full-scale production faster, easier and with less risk
EU CRA compliance	On-going maintenance and feature update by Nordic ensuring adherence to EU CRA

Offer

nRF CLOUD powered by Memfault

Lifetime FOTA

Starting from

$1.00 / device

The industry's only lifetime FOTA coverage pricing exclusive for teams building on Nordic ultra-low power wireless solutions.

- Advertised pricing based on 10Ku
- Lifetime FOTA coverage
- Volume discounts available
- Terms apply

Prices listed are subject to change without notice and are valid as of the date of publication.
All pricing is exclusive of applicable taxes, fees, or third-party charges unless otherwise stated.

NORDIC SEMICONDUCTOR

Nordic product overview

Cloud support across all our wireless connectivity solutions

Short-Range	Cellular IoT	Wi-Fi 6 IoT	Power Management
nRF54 SERIES	nRF91 SERIES	nRF70 SERIES	nPM FAMILY
Bluetooth, THREAD, matter, zigbee	LTE-M, NB-IoT, GNSS	Wi Fi, matter	
nRF CLOUD powered by Memfault	nRF CLOUD powered by Memfault	nRF CLOUD powered by Memfault	

Nordic's power management platform

Enabling highly integrated PMIC products

nPM FAMILY

Battery management

- Battery charger
- Fuel gauge

System management features

Recovery and protection
- Boot failure recovery
- Hard reset
- Watchdog timer
- Power loss warning

Energy conservation
- Ship mode
- Hibernate mode

Interfaces
- SDA CLK Two-wire I/F
- USB charge I/F
- LED driver
- GPIO

Voltage regulators

- Buck converter
- Boost converter
- LDO Low Drop-Out regulator

PMIC product portfolio + roadmap

Advanced battery management in compact packages

In production
In sampling
In design

nPM1100		**Li-Ion & LiPo Charger + DC/DC**	GPIO controlled, basic system features, 400mA max. charge current
nPM1300		**Li-Ion, LiPo & LiFePO4 Charger + DC/DC**	TWI controlled, FG + adv. system features, 800mA charge current
nPM1304		**Li-Ion, LiPo & LiFePO4 Charger + DC/DC**	Optimized for small batteries, max. charge current 100mA
nPM1012	Sampling now	**Li-Ion, LiPo, LiFePO4 & LiC Charger + DC/DC**	For small batteries, compatible with NFC wireless chargers
nPM12 Series	Sampling Q4'26	**Li-Ion, LiPo, LiFePO4 & LiC Charger + DC/DC**	For medium size batteries, compatible with NFC wireless chargers
nPM2100		**Primary cells (AA, LR44, CR type) DC/DC boost + FG**	Boost output of 150mA
nPM6001		**Multi-rail PMIC**	

nRF Connect SDK

Software Development Kit

Nordic nRF Connect

nRF Connect

nRF Connect SDK
- Samples and Applications
- Middleware
- RTOS
- Libraries
- Hardware drivers

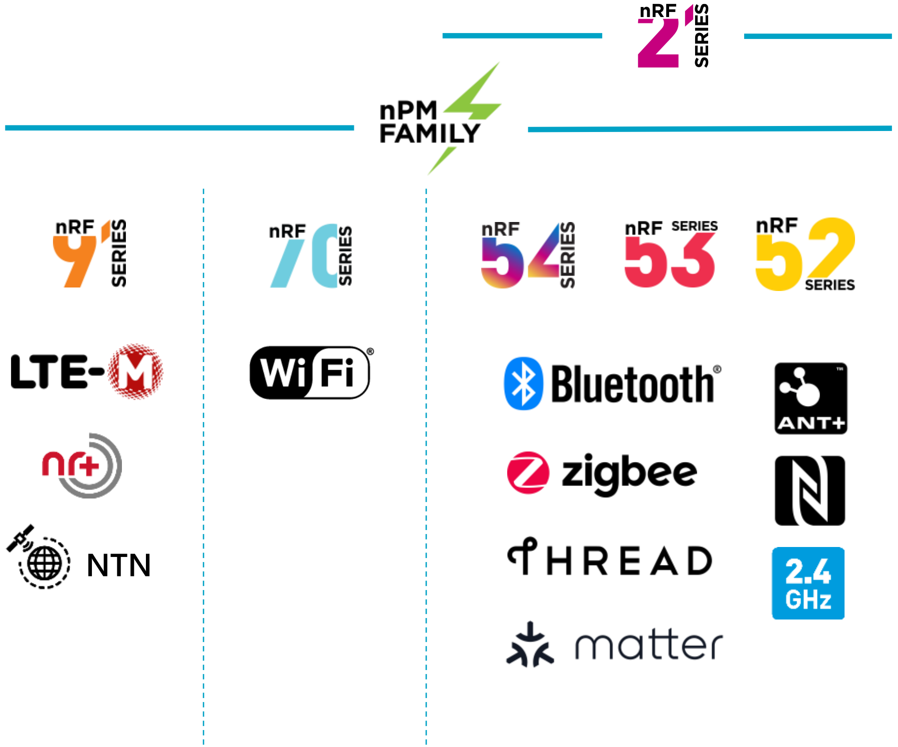
nRF Connect for Desktop
- Toolchain Manager
- Bluetooth Low Energy
- Programmer
- LTE Link Monitor
- Power Profiler
- Trace Collector

nRF Connect for VS Code
- nRF Terminal
- nRF Debug
- Memory Report
- Command-line (CLI) and GUI interfaces
- Create new board wizard

Mobile Applications
- nRF Connect for Mobile
- nRF Mesh
- nRF Toolbox
- Nordic Thingy
- nRF Cloud Gateway

nRF Connect SDK

- One code base and toolchain for all current Nordic products
- Covers all Nordic supported technologies such as Bluetooth, Bluetooth Mesh, IEEE 802.15.4 (Thread + Zigbee), ESB, NFC, ANT, Wi-Fi, LTE-M, NB-IoT, GNSS, DECT NR+
- Access to latest certified Bluetooth stack
  - Bluetooth Core v6.2, Mesh v1.1.1
  - (Channel sounding, SCI & more)



nRF Connect SDK Code Base

- **Code base consists of several integrated repositories**
  - Code coming from Nordic (blue)
  - Code coming from Open Source (OS) repositories (purple)
    - (Partially Nordic as well, as Nordic contributes much into Zephyr & MCUBoot and related OS projects)

- **Contains an RTOS, application code, connectivity protocols, wireless stacks, peripheral drivers & more**

nRF Connect SDK Repositories

- nRF: Application & connectivity protocols
- nrfxlib: compiled libraries & pre-certified stacks
- nrfx: peripheral drivers
- Zephyr: RTOS & middleware (OS)
- MCUboot: Secure Bootloader (OS)
- Other repositories
  - Trusted Firmware-M, Matter, etc.

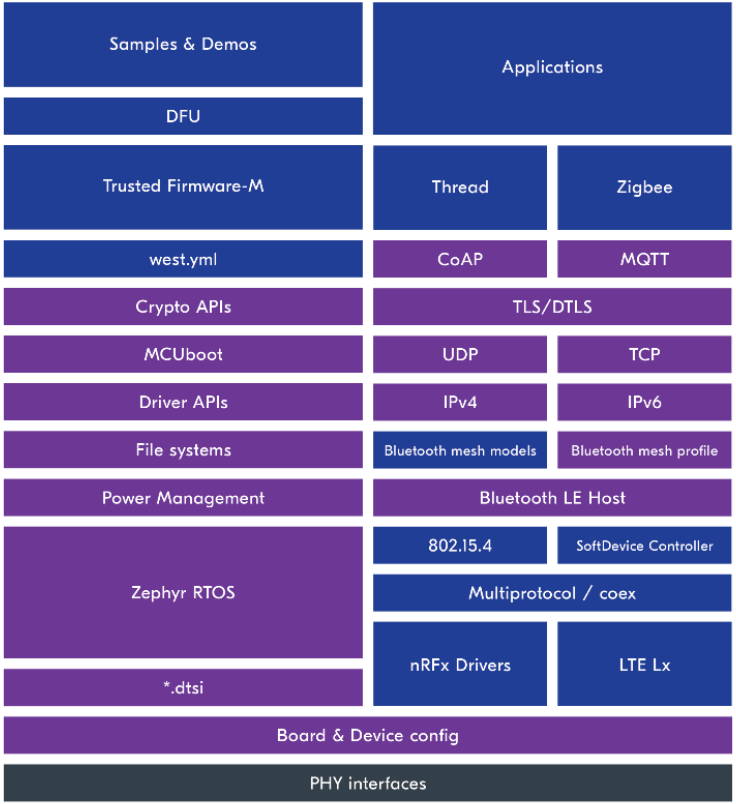
Nordic and Open Source (OS) code

nRF Connect SDK

Code Base Details

- **Zephyr includes a lot of fundamental pieces of code and functionality**
  - Zephyr peripheral drivers
  - Middleware & messaging protocols (IP, HTTP, MQTT, CoAP, Modbus, …)
  - File system
  - Device drivers (e.g. sensors)

- Allows developers to focus on the components specific to their designs while having ready-to-use components for trivial functionality



Why the Zephyr project?

- **Independent open-source technology project**
  - Governed under the Linux Foundation
  - Contributions made by over 1000 industry embedded experts
- **Nordic has been a member & contributor since 2016**
- **Scalability & Design Flexibility for IoT devices**
  - Allows for very small configurations on constrained devices (min. 8 kB e.g. simple LED blinking application)
  - Scalable to powerful, feature-rich, configurations for large memory, high-processing power devices
  - Cross-Architecture (Arm, RISC-V, MIPS, x86, Xtensa & more)

Platinum Members

Activity of the Zephyr project

Zephyr

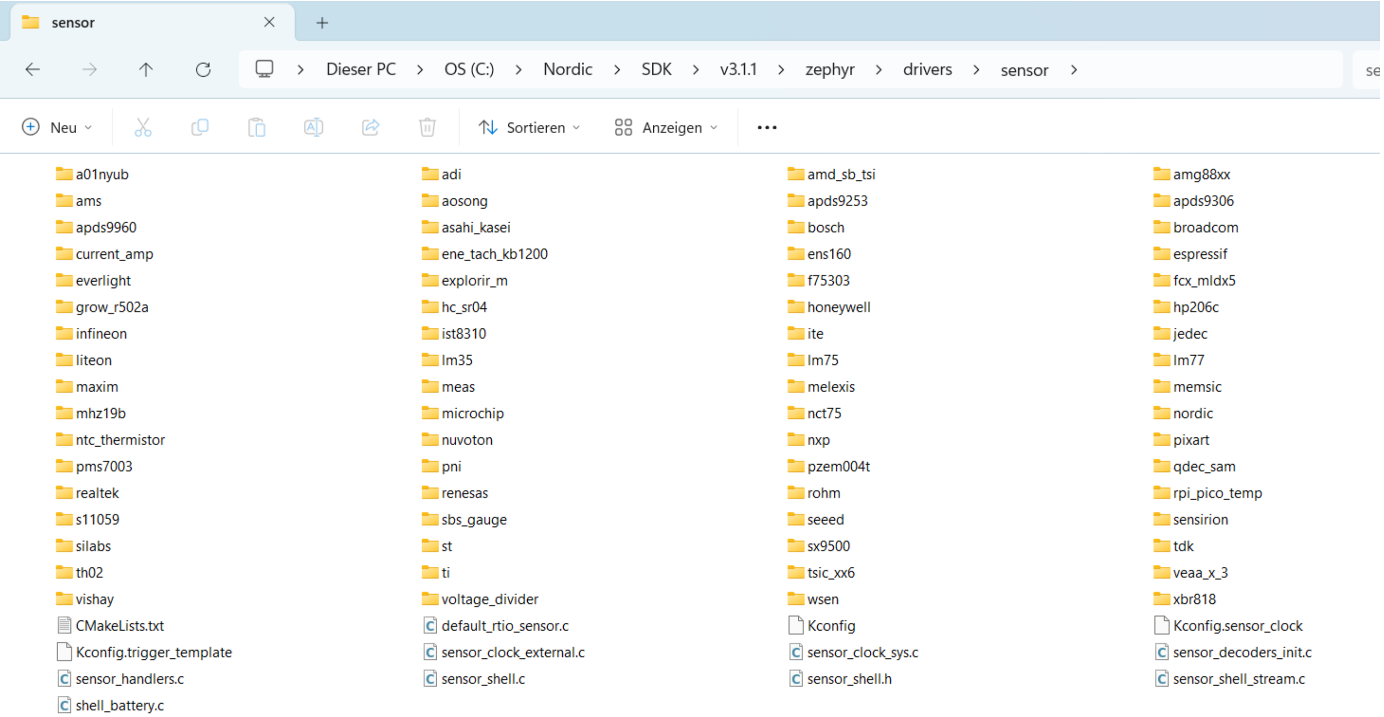
Operating System	First Commit	Controls Commits	Declared License	Total Contributors	Contributors in last month	Total Commits	Commits in last month
Zephyr	2014/11	**community**	**Apache-2.0**	**2568**	**353**	**115,236**	**1,732**
nuttX	2007/?	community	BSD-variant → Apache-2.0	620	43	58,459	262
RIOT	2010/09	community	LGPL-2.1	373	16	47,709	110
RT-Thread	2009/06	community	GPL-2.0 → Apache-2.0	745	19	17,107	72
Tizen RT	2015/04	Samsung	BSD-variant → Apache-2.0	208	16	11,712	55
Ariel-OS	2020/08	community	Apache-2.0 OR MIT	18	7	2,582	49
myNewt	2015/06	community	Apache-2.0	135	3	11,200	13
SeL4	2014/07	community	GPLv2 AND BSD-2-Clause	115	4	4,694	8
FreeRTOS	2004/07	Richard Barry	GPL-2.0 w/ FreeRTOS → MIT	215	6	3,591	6
Contiki-NG	2017/10	community	BSD-3-Clause	220	2	17,997	2
ThreadX	2020/05	MSFT → community	MSL → MIT	22	0	225	0

Data extracted on 2025-05-15 from github

21

Leverage code that is fully integrated

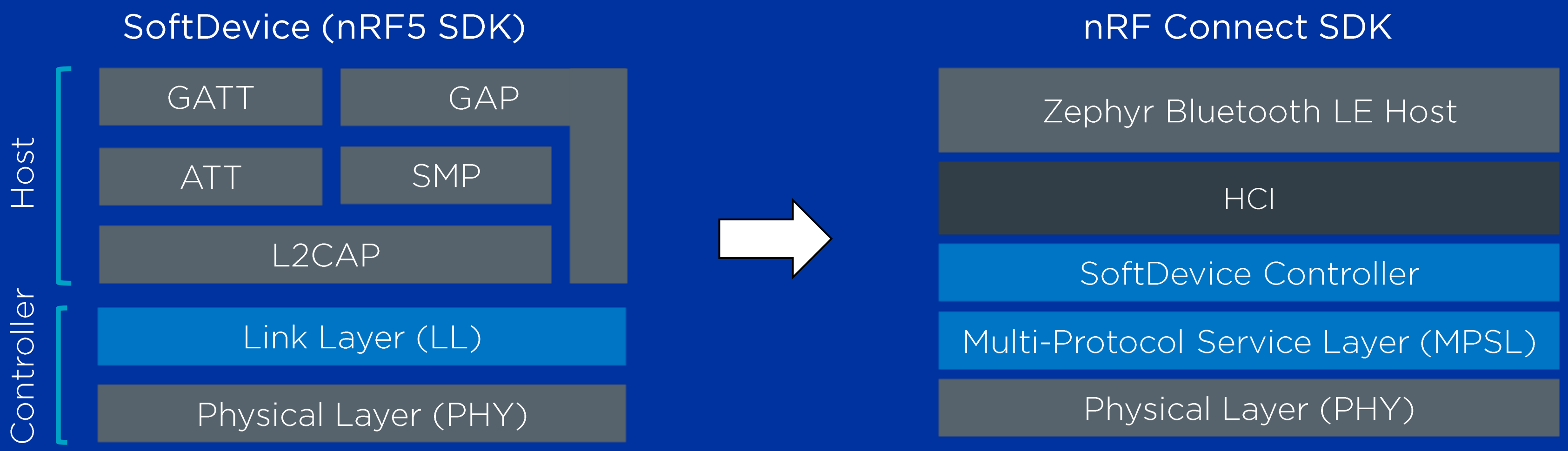
Integrated sensor drivers



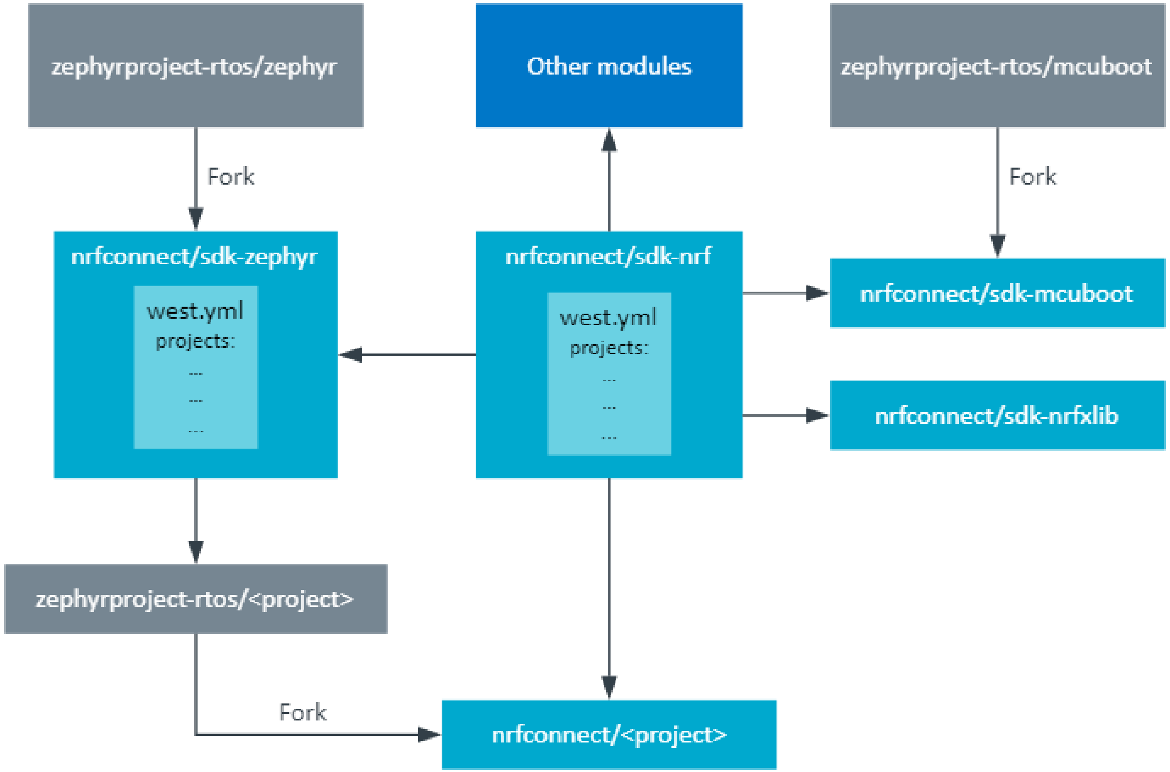
Nordic SoftDevice Controller

Industry proven Bluetooth LE stack

- Nordic ported the SoftDevice Controller to nRF Connect SDK
- The SoftDevice controller is the default Bluetooth LE controller for Bluetooth samples on Nordic HW

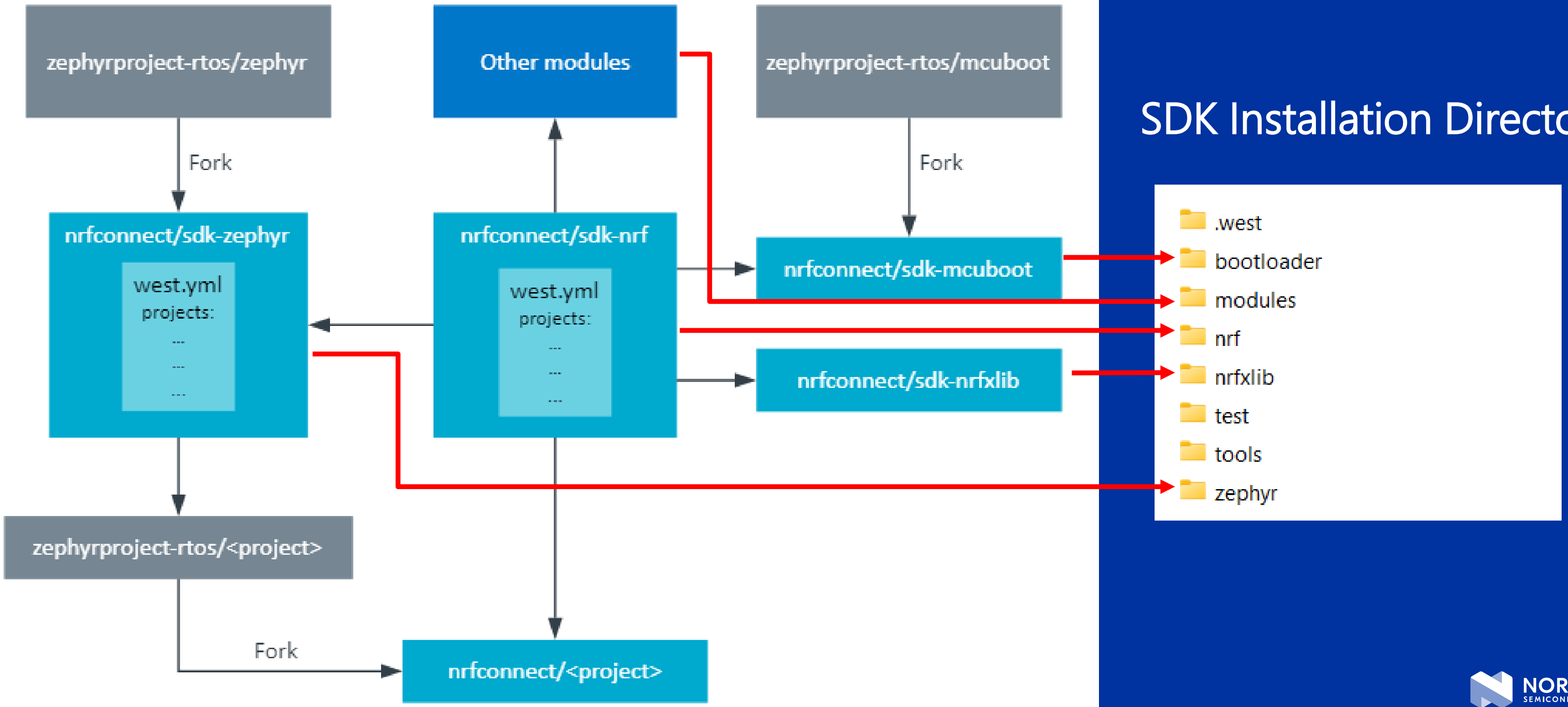


Repository Structure



- Star topology, sdk-nrf is the central / main repository of the SDK
  - west.yml is the manifest
- Fork from Zephyr vanilla repositories
  - sdk-zephyr, hal_nordic, mcuboot, little_fs, mbed_tls
- Nordic private repositories
  - nrfxlib, homekit, find-my,

Repository Structure

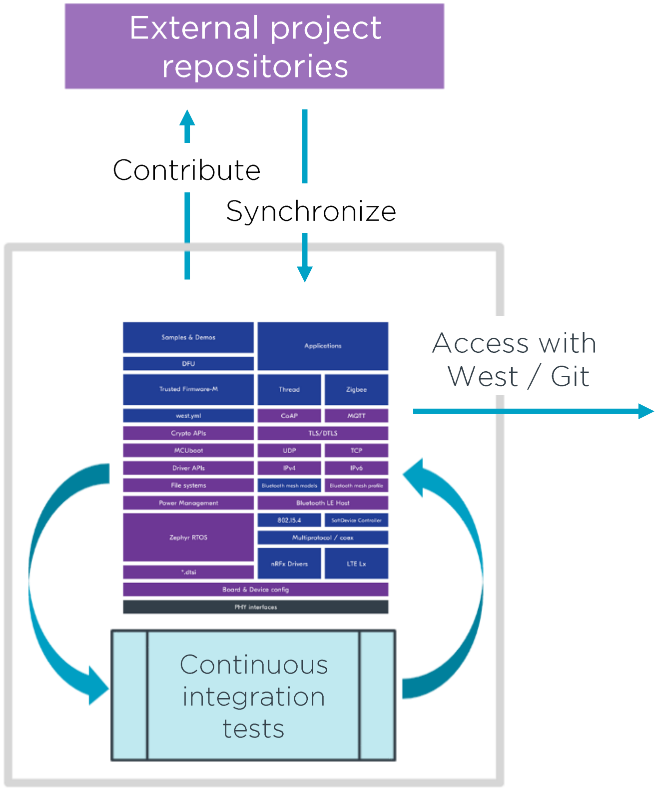


SDK Installation Directory

nRF Connect SDK Releases

- **nRF Connect SDK is a single platform**
  - All source code is distributed by Nordic
  - Includes open-source code from external projects
  - Nordic contributes to, and synchronizes with external projects

- **Nordic runs integration test on all source code and manages configuration compatibilities**
  - Once completed, specific source code tags are released as SDK version



nRF Connect SDK

Toolchain

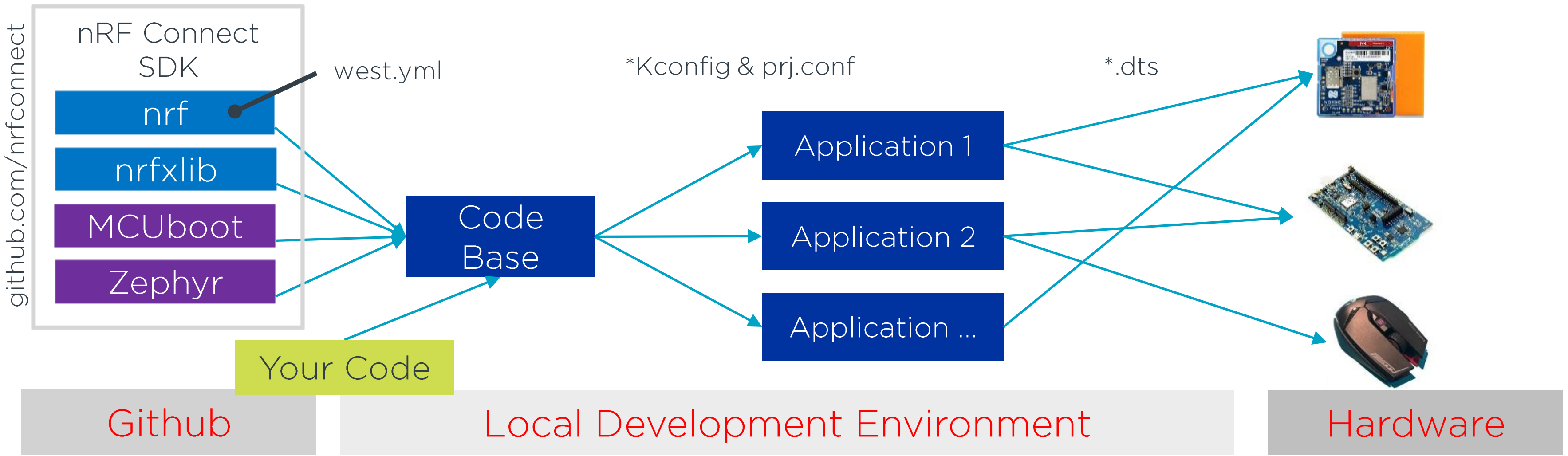
Zephyr toolchain overview

West
Multi-repository management tool

Kconfig
Source module / feature configuration for compile

Device Tree
Target Board / Device description

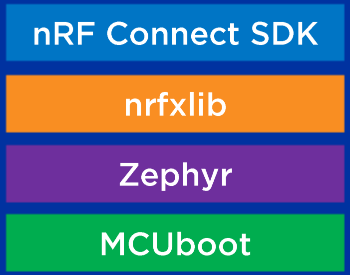
Clone / update → Configure features → Configure target

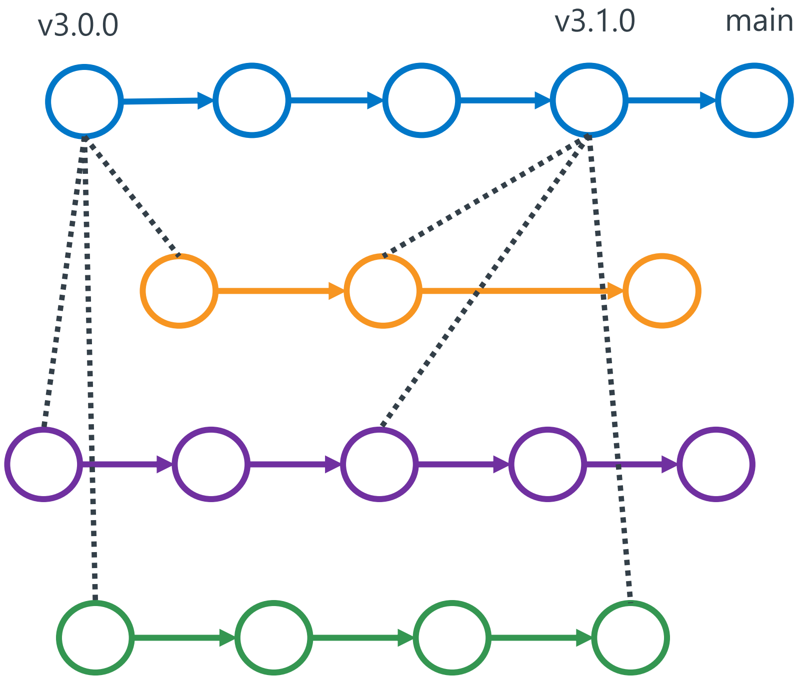


West Tool

- **Used as top level command interface**
  - west build, west flash, west debug etc.

- **Used for multi-repository management**
  - Reads the SDK manifest file (west.yml)
  - Runs git underneath



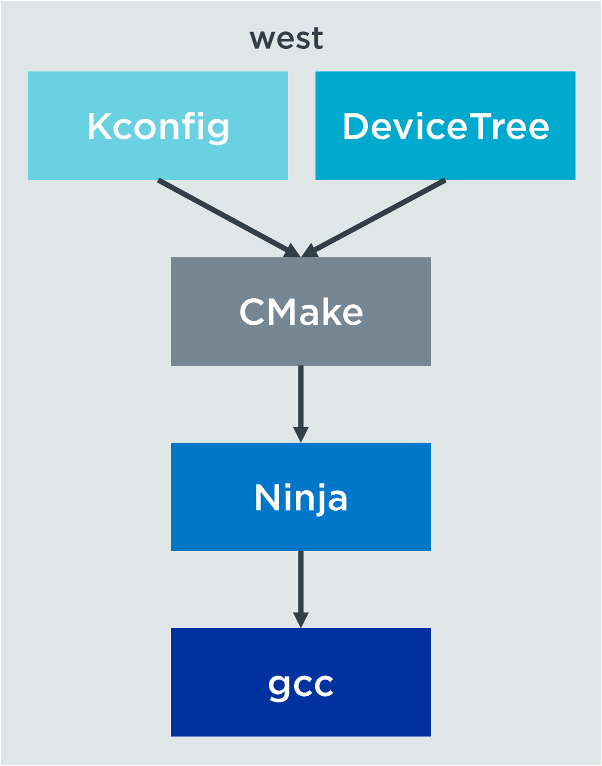


Kconfig & DeviceTree

Adjust the build without code changes

- **Kconfig**
  - Manages system features and software libraries
  - Generates C definitions to configure the system and include libraries without changing the source code (e.g. BLE, sensors, RTOS services)

- **DeviceTree**
  - Describes hardware and pin layout
  - Stored in .dts or .dtsi file format
  - Allows for modifications via an overlay file

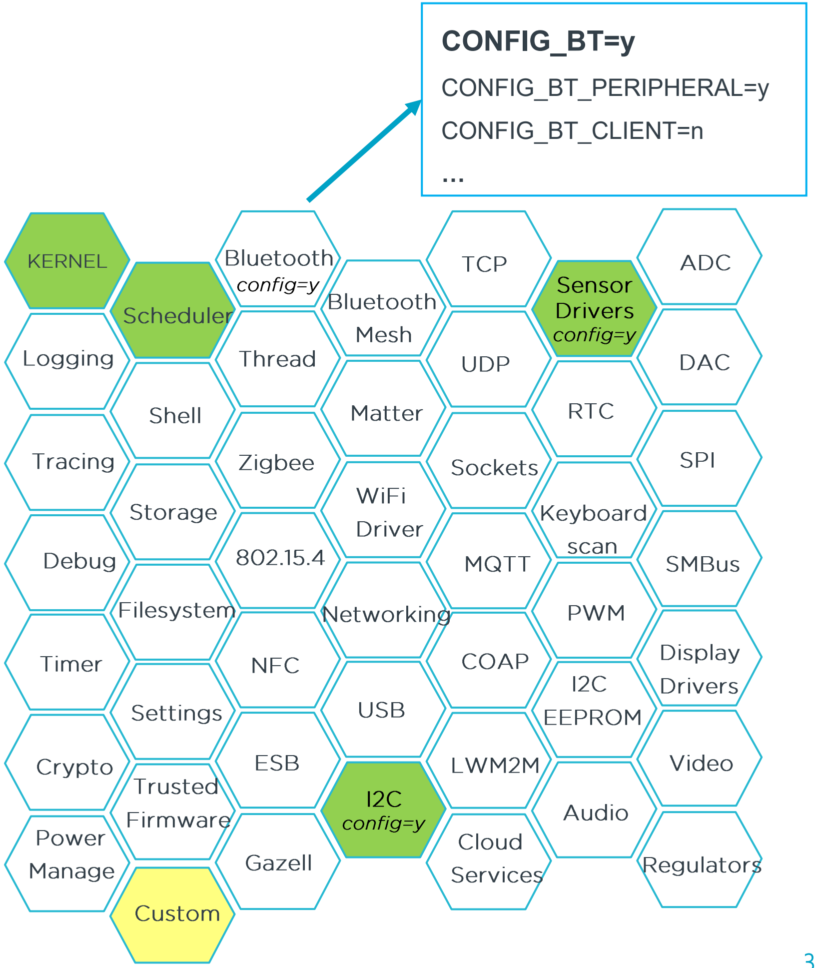


Project Kconfig

Zephyr Kernel Configuration

- **Zephyr kernel and sub systems can be configured at build time**
  - Add software modules to your project using the project configuration file (prj.conf)
  - If a software module was added, further CONFIG symbols appear that allow you to configure the module

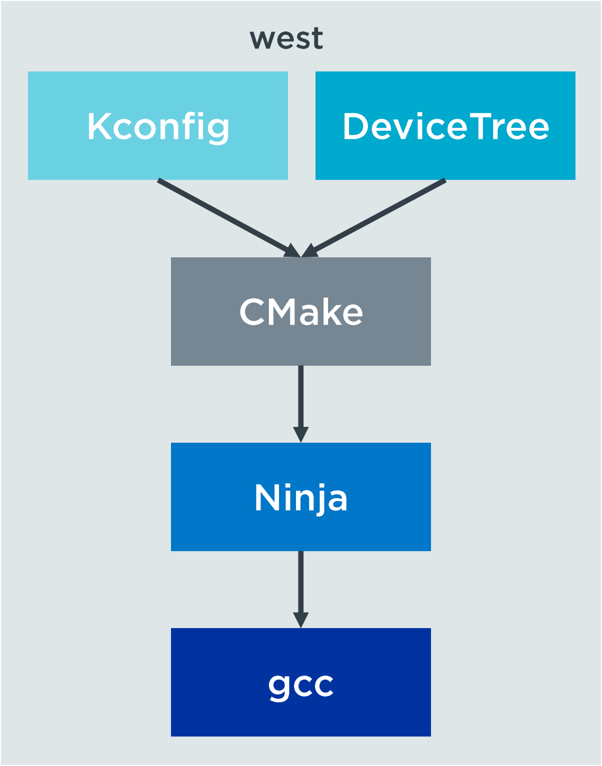
- **Custom Kconfig definitions are possible**



CMake, Ninja, gcc

Powerful & open-source build tools

- **CMake**
  - Cross-platform build generator
- **Ninja**
  - Build executor, similar to make
  - Faster than make when performing incremental builds
- **gcc**
  - Compiler and linker



Visual Studio Code as IDE

+ nRF Connect Extension Pack

Known platform and feature-rich addons:

- CLI and GUI interface
- DeviceTree Visual Editor
- SDK/Toolchain management
- nRF Debug
- Memory Report
- Built-in Serial Terminal
- nRF Kconfig GUI

Rich set of tutorial videos

Interactive Online Learning Platform

A learning path from Fundamentals to Intermediate

DevAcademy

nRF Connect SDK Fundamentals

Bluetooth LE Fundamentals

nRF Connect SDK Intermediate

THANK YOU

Your low power wireless connectivity solutions partner